

Cambridge (CIE) IGCSE Physics
Paper 1: Multiple Choice (Set C)
Core

Saturday 26 April 2025

Morning (Time: 0 hours 45 minutes)

Total marks

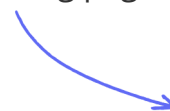
/ 40

Instructions

- Try to complete this mock exam paper in one sitting, under exam conditions. Use all the time available and check your answers to each question at the end before submitting.
- Remember this is PRACTICE. Mistakes are fine and will help you improve in time for the real exam - just do your best.
- You may use a calculator.
- Take the weight of 1.0 kg to be 9.8N (acceleration of free fall = 9.8m/s^2).

Scan here to mark your mock exam

or visit the mock exams landing page for this course



- 1 A student is timing the rate of cooling of a beaker of water. On a repeat run of the investigation, the student forgets to zero the stopwatch.

The readings on the stopwatch at the start time and the end time are shown in Fig. 1.1

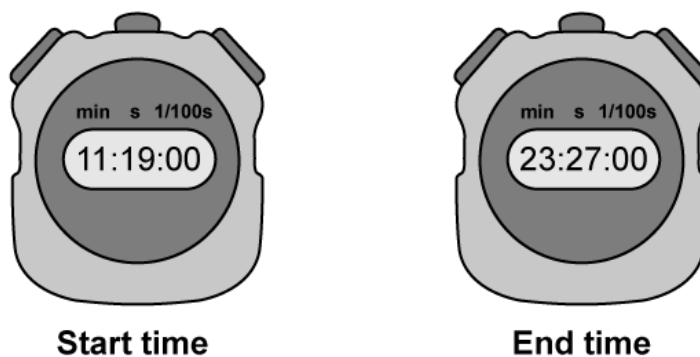


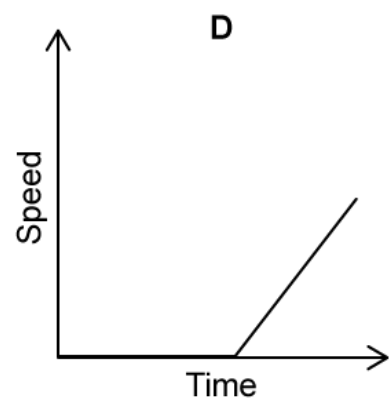
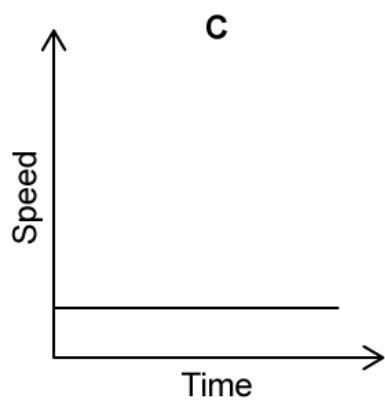
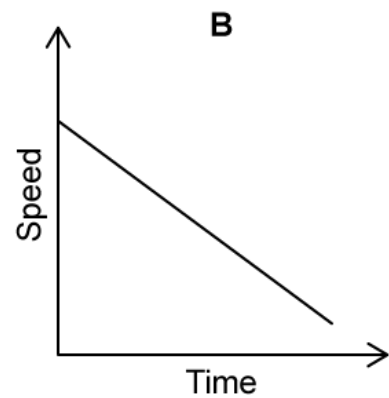
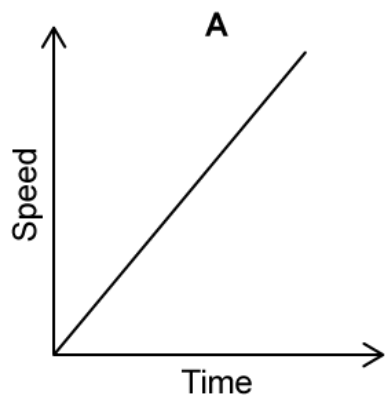
Fig. 1.1

Calculate how long it took the beaker to cool.

- A. 11 minutes
- B. 11 minutes and 8 seconds
- C. 12 minutes
- D. 12 minutes and 8 seconds

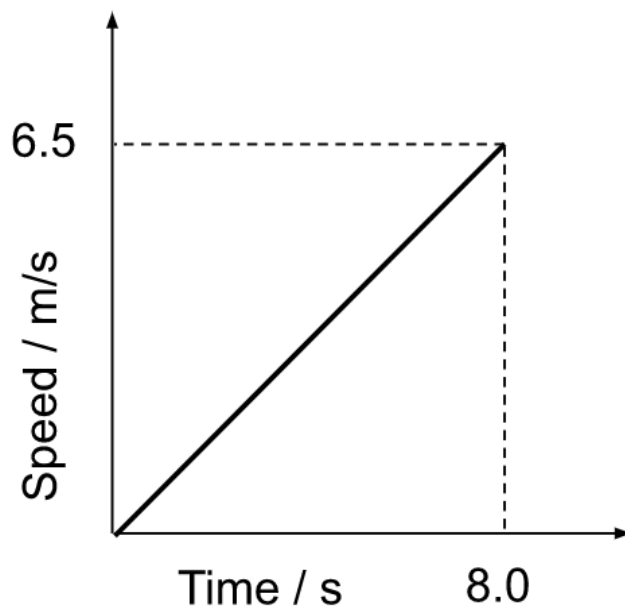
(1 mark)

2 Which speed-time graph shows an object moving with a constant speed?



(1 mark)

3 The graph shows the journey undertaken by a car.



Which equation correctly gives the distance travelled by the car?

A. $\frac{6.5 \times 8.0}{2} = 26\text{m}$

B. $6.5 \times 8.0 = 52 \text{ m}$

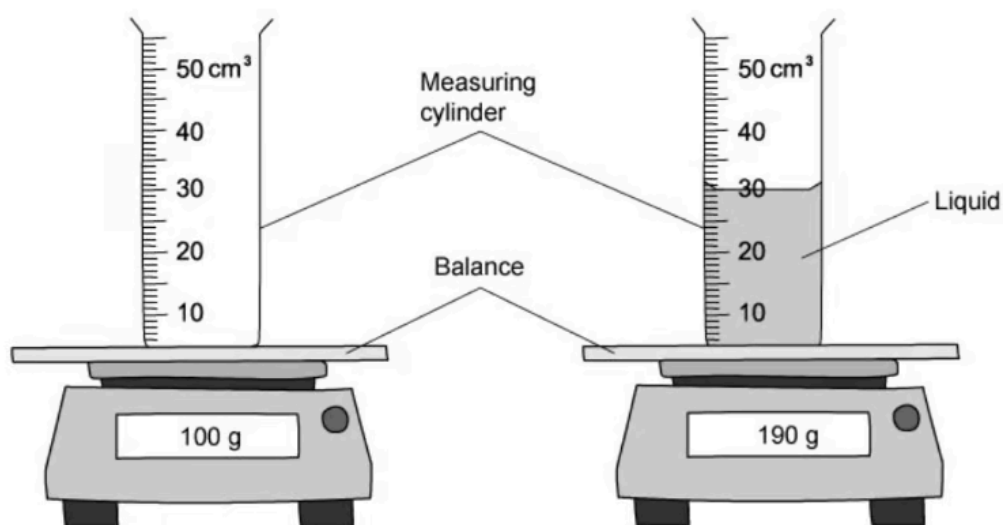
C. $\frac{6.5}{8.0} = 0.81 \text{ m}$

D. $\frac{8.0}{6.5} = 1.2 \text{ m}$

(1 mark)

4 The diagram shows a measuring cylinder containing a liquid , and that same measuring

cylinder when it is empty.



What is the density of the liquid?

- A.** 6.30 g/cm^3
- B.** 3.00 g/cm^3
- C.** 0.33 g/cm^3
- D.** 0.16 g/cm^3

(1 mark)

- 5** A teacher has a hot cup of tea.

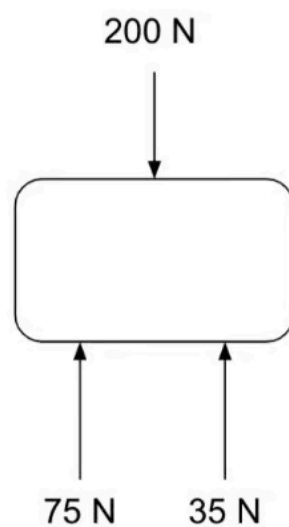
Some of the liquid in the cup evaporates.

Choose the line from the table below that correctly describes what happens to the mass and the weight of the tea in the cup.

	Mass	Weight
A	increases	decreases
B	increases	increases
C	decreases	decreases
D	decreases	increases

(1 mark)

6 An object with three forces acting on it is shown below.



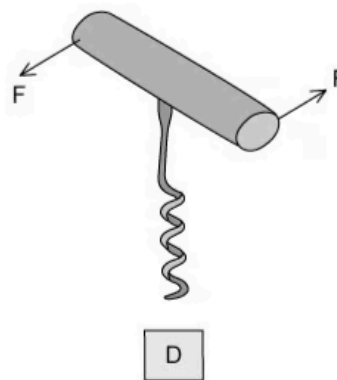
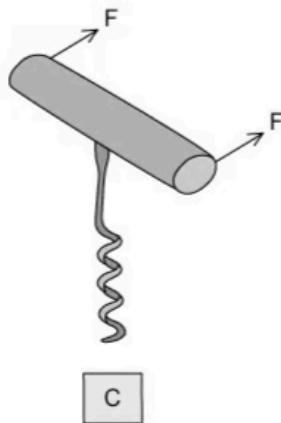
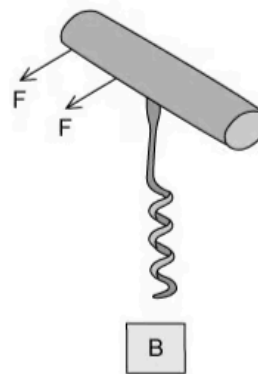
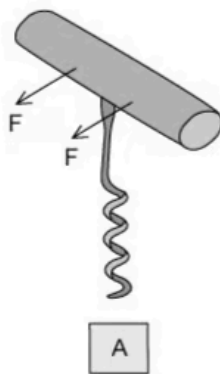
What is the resultant force acting on the object?

- A.** 90 N downwards
- B.** 110 N upwards
- C.** 200 N downwards
- D.** 90 N upwards

(1 mark)

7 A corkscrew is used to open a bottle of wine.

Two forces are applied at each of the positions shown below.



Which pair of forces produces the greatest turning force?

(1 mark)

8 In which of the following situations would no resultant force be required?

- A. A motorbike travelling round a bend at a constant speed.
- B. A rocket accelerating straight upwards into space.
- C. A ship sailing at a constant speed across the ocean.
- D. A Formula One car accelerating off the starting grid.

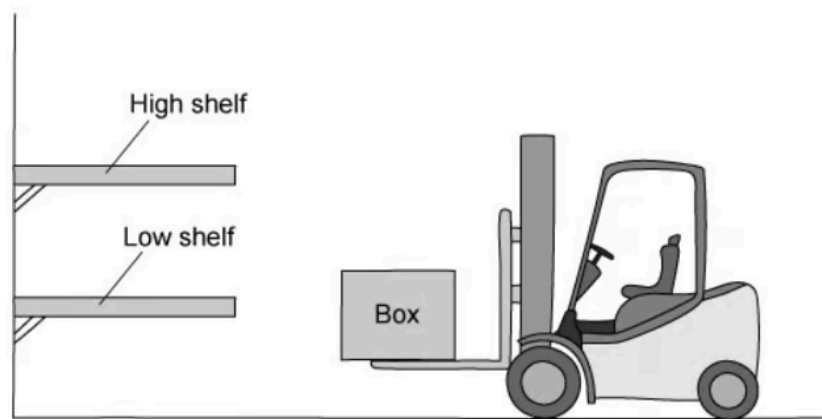
(1 mark)

9 Which row correctly matches an energy store with an example of that store?

	Energy Store	Example
A.	kinetic	a car moving at 70 km / h
B.	elastic	a box sitting high on a shelf
C.	magnetic	compression of springs in a car's suspension
D.	chemical	the nuclei of uranium atoms

(1 mark)

10 A fork lift truck lifts a heavy box onto a shelf.

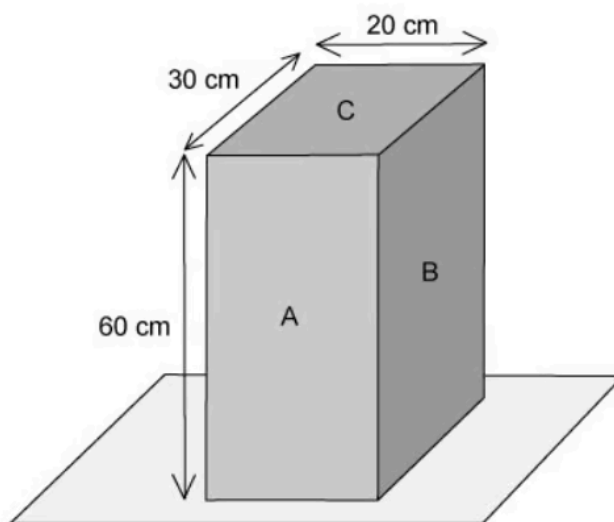


Which action requires the greatest power output from the fork lift truck.

- A. Lifting the box from the ground to the high shelf slowly
- B. Lifting the box from the ground to the high shelf quickly
- C. Lifting the box from the low shelf to the high shelf quickly
- D. Lifting the box from the low shelf to the high shelf slowly

(1 mark)

- 11 A wooden block rests on a table.



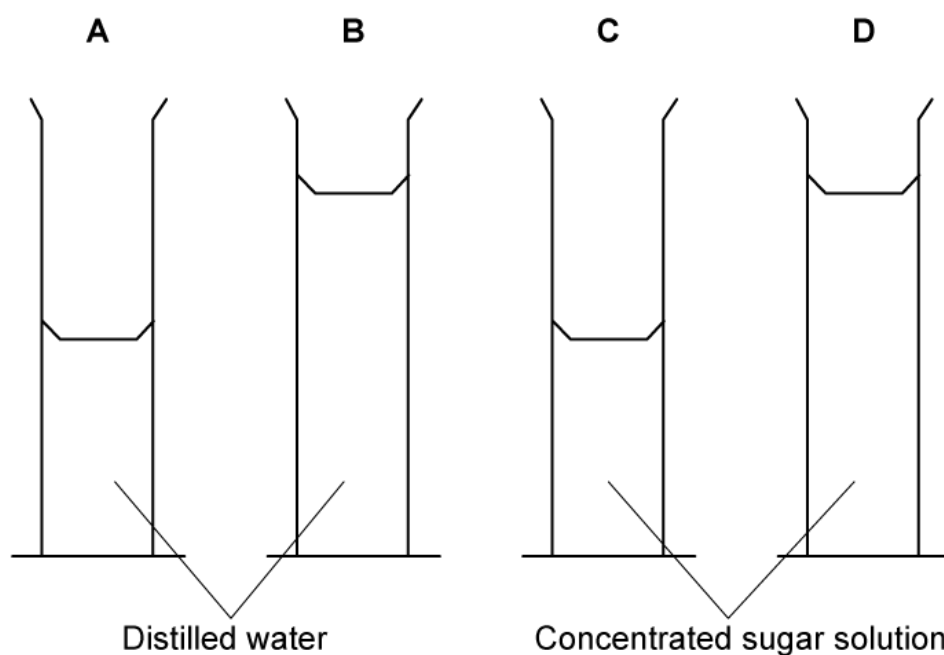
On which surface should the block be laid to produce the largest pressure on the table?

- A. A
- B. B
- C. C
- D. Any side, they will all produce the same pressure

(1 mark)

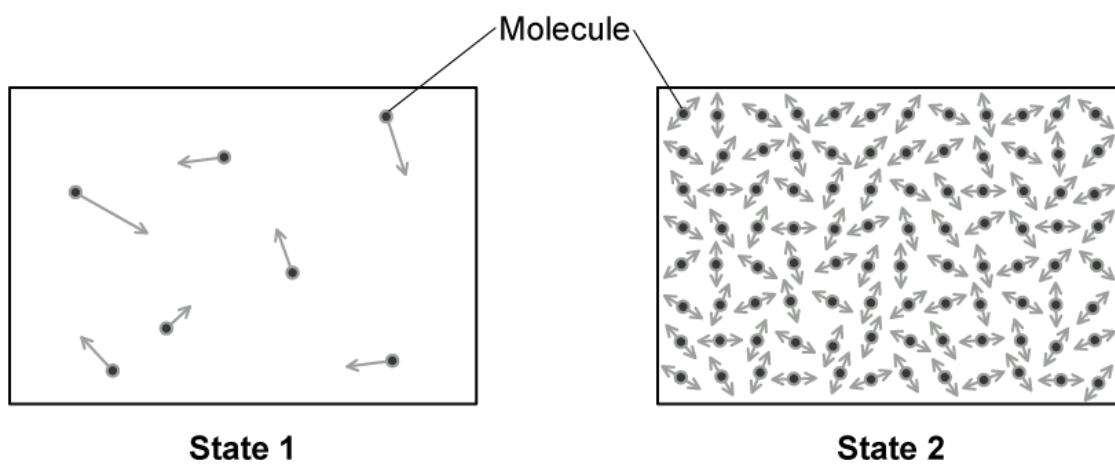
- 12 The diagram shows four measuring identical cylinders containing either distilled water or concentrated sugar solution.

Which measuring cylinder has the least pressure at the base, due to the liquid?



(1 mark)

- 13 The diagrams represent the molecules in two different states of matter. The arrows show the motion of the molecules.



Which state represents a solid?

- A. state 1
- B. neither

- C. both
- D. state 2

(1 mark)

- 14 A beaker of water, at its boiling point of 100°C , is heated so that the liquid starts to become a gas, also at 100°C .

Which row of the table correctly compares the liquid phase with the gas phase under these conditions?

	Average distance between the particles	Average speed of the particles
A	the same in the liquid and the gas	the same in the liquid and the gas
B	greater in the liquid	greater in the gas
C	greater in the gas	greater in the liquid
D	greater in the gas	the same in the liquid and the gas

(1 mark)

- 15 An iron cube is heated by a Bunsen burner.

Which of its properties decreases?

- A. Mass
- B. Density
- C. Volume
- D. Internal energy

(1 mark)

- 16** Internal energy is affected by two properties of the particles which make up a substance.

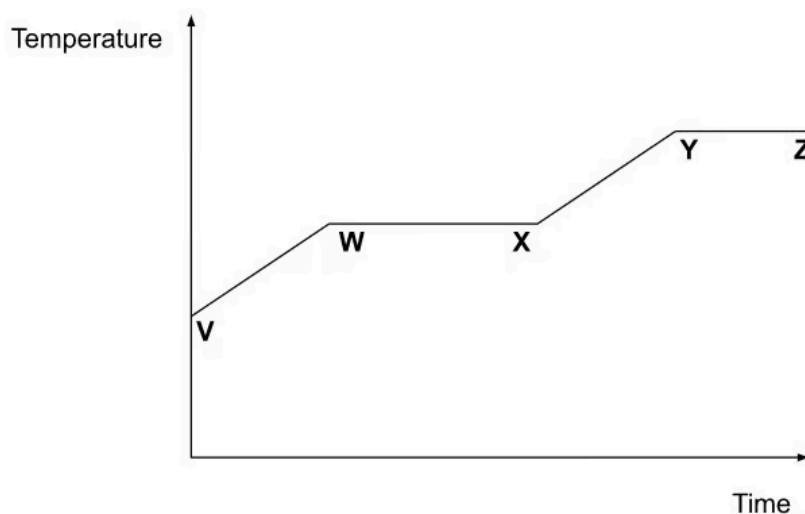
Identify the two properties.

- A.** Motion and position
- B.** Acceleration and mass
- C.** Specific heat capacity and volume
- D.** State of matter and potential energy

(1 mark)

- 17** A substance, which is initially in its solid form, is heated at a constant rate.

The graph shows how its temperature changes with time.



Between which points is the substance partly a liquid and partly a gas?

- A.** V and W
- B.** W and X
- C.** X and Y
- D.** Y and Z

(1 mark)

18 Which of the following statements about convection is correct?

Warm air rises because:

- A.** The air molecules expand and become less dense.
- B.** The air molecules move further apart, making the air less dense.
- C.** The air molecules contract, making the air less dense.
- D.** The air molecules move closer together, making the air less dense.

(1 mark)

19 Two cars sit side by side, a black car and a white car. It is Summer.



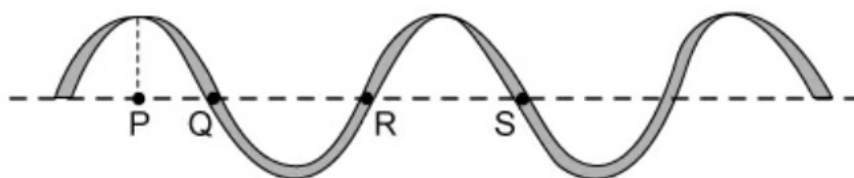
During the day both cars absorb infrared radiation from the Sun, and during the night, both cars lose energy by emitting infrared radiation.

Which car heats up more quickly during the day, and which car cools down more quickly at night?

	Car which heats up more quickly during the day	Car which cools down more quickly at night
A	black	black
B	black	white
C	white	black
D	white	white

(1 mark)

20 Which two points on the wave are exactly one wavelength apart?



- A. P and S
- B. Q and R
- C. Q and S
- D. P and R

(1 mark)

21 The following table shows some examples of waves

Which row correctly lists the nature of each of the wave types?

	Sound waves	Infrared waves	Red light waves
A	transverse	longitudinal	transverse
B	longitudinal	transverse	transverse
C	longitudinal	longitudinal	transverse
D	transverse	transverse	longitudinal

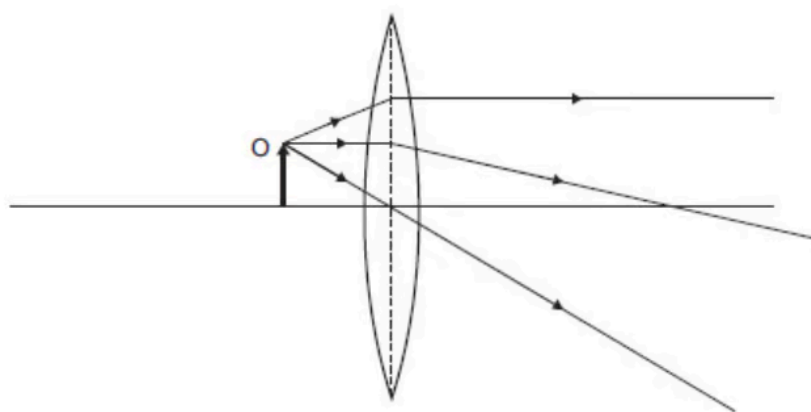
(1 mark)

22 When light passes from a denser material into a less dense material

- A.** light slows down and bends toward the normal
- B.** light slows down and bends away from the normal
- C.** light speeds up and bends toward the normal
- D.** light speeds up and bends away from the normal

(1 mark)

23 The diagram shows an object positioned close to a thin converging lens (inside the focal distance). Three light rays are shown leaving the object and passing through the lens.



Which row in the table correctly describes the properties of the image?

	location	size	real / virtual
A	to the left of O	reduced	real
B	to the right of O	enlarged	virtual
C	to the left of O	enlarged	virtual
D	at O	the same as O	real

(1 mark)

- 24** The table below shows the entire electromagnetic spectrum. It runs in order of increasing wavelength.

Which region is microwaves?

γ-rays	A	B	C	infrared	D	radio waves
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(1 mark)

- 25** Which type of electromagnetic wave is used in remote controls?

- A.** Microwaves
- B.** Visible light
- C.** Ultraviolet
- D.** Infrared

(1 mark)

- 26** Which of the lists below are metals that are ferrous?

- A.** aluminium and iron
- B.** steel and iron
- C.** steel and lead
- D.** tin and iron

(1 mark)

27 A student makes a series of statements about a permanent bar magnet.

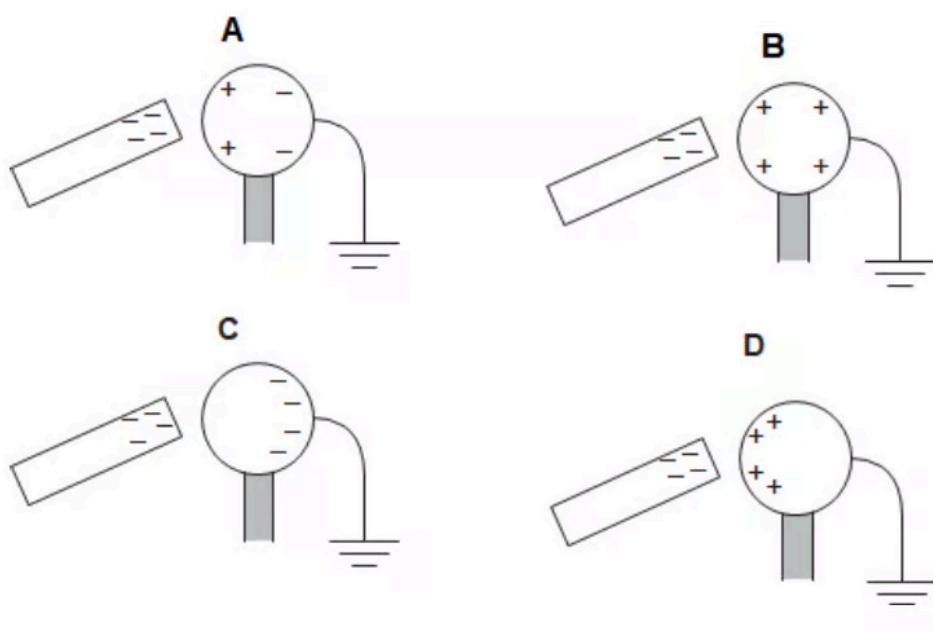
Which of them is true?

- A.** The North pole of a permanent bar magnet will repel the North pole of another magnet.
- B.** A permanent bar magnet must be made from a soft magnetic material.
- C.** A permanent bar magnet can repel a magnetic material.
- D.** The field lines of a permanent bar magnet cross each other where the field is strongest.

(1 mark)

28 A student rubs an acetate rod with a cloth, giving it a negative charge. She then holds it near an earthed, conducting sphere as shown in the diagrams below.

Which diagram shows the correct distribution of charges on the conducting sphere?



(1 mark)

29 What is the unit of electromotive force?

- A.** Joule
- B.** Volt
- C.** Ampere
- D.** Watt

(1 mark)

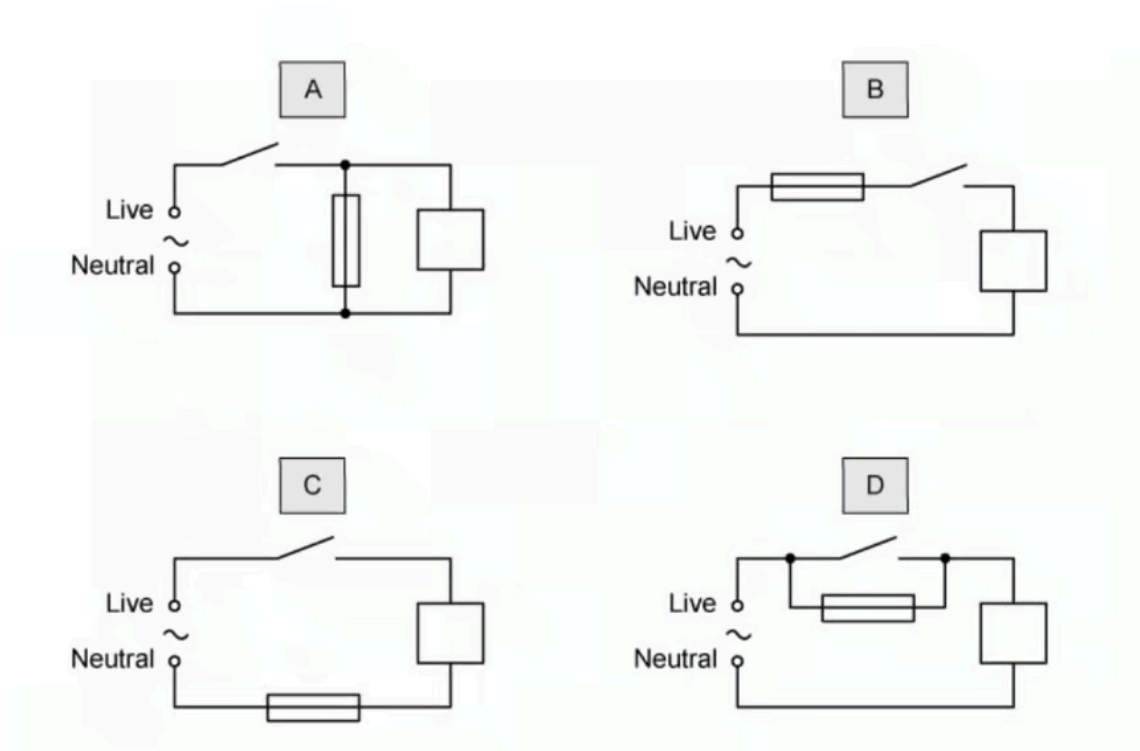
30 A student connects a 6 V power supply to a 3 Ω resistor. The resistor is left connected to the power supply for 1 minute.

How much power is dissipated by the resistor?

- A.** 2 W
- B.** 12 W
- C.** 720 J
- D.** 18 W

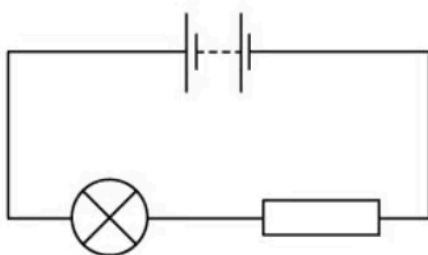
(1 mark)

31 Which of the circuit diagrams above shows a fuse which has been correctly connected?



(1 mark)

32 A book shop owner sets up the circuit below, but they find that the lamp is too dim.

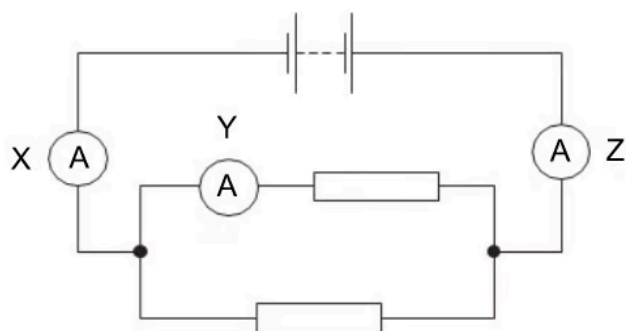


What change can be made to the circuit to increase the brightness of the bulb?

- A.** Putting another resistor in parallel with the battery.
- B.** Putting a resistor in parallel with the bulb.
- C.** Putting another resistor in series with the resistor.
- D.** Putting another resistor in parallel with the resistor.

(1 mark)

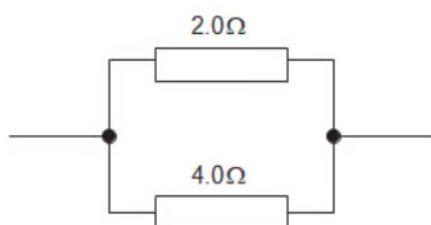
33 Which of the following statements about the current in ammeters **X**, **Y** and **Z** is correct?



- A.** The current through X equals the current through Y + the current through Z.
- B.** The current through X is equal to the current through Z.
- C.** The current through X is equal to the current through Y.
- D.** The current through Y is greater than the current through Z.

(1 mark)

34 The diagram shows two resistors connected in parallel.



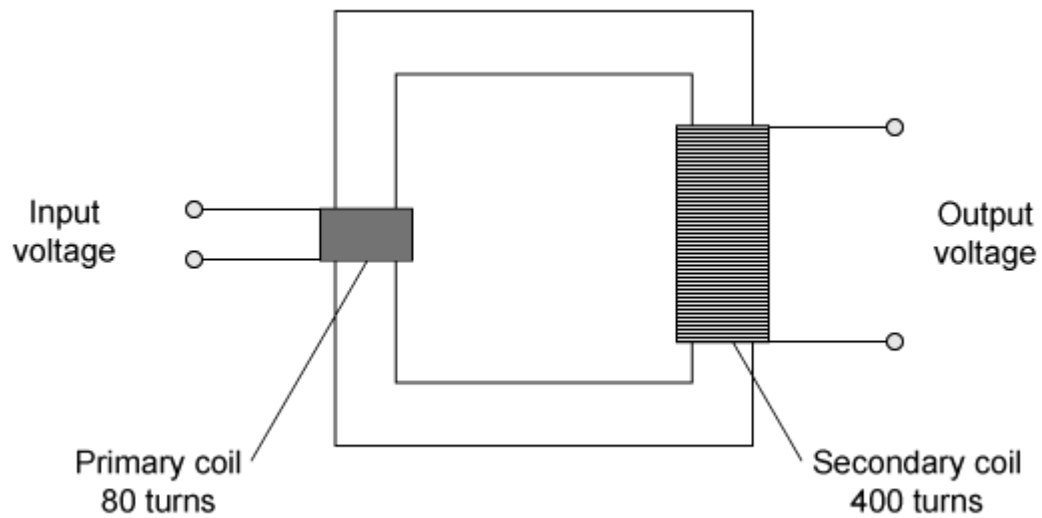
The total resistance of this combination will be:

- A.** The sum of $2.0\ \Omega$ and $4.0\ \Omega$
- B.** Greater than $4.0\ \Omega$
- C.** Between $2.0\ \Omega$ and $4.0\ \Omega$
- D.** Less than $2.0\ \Omega$

(1 mark)

35 Transformers are used to change the voltage of a power supply.

The diagram below shows a simple transformer.



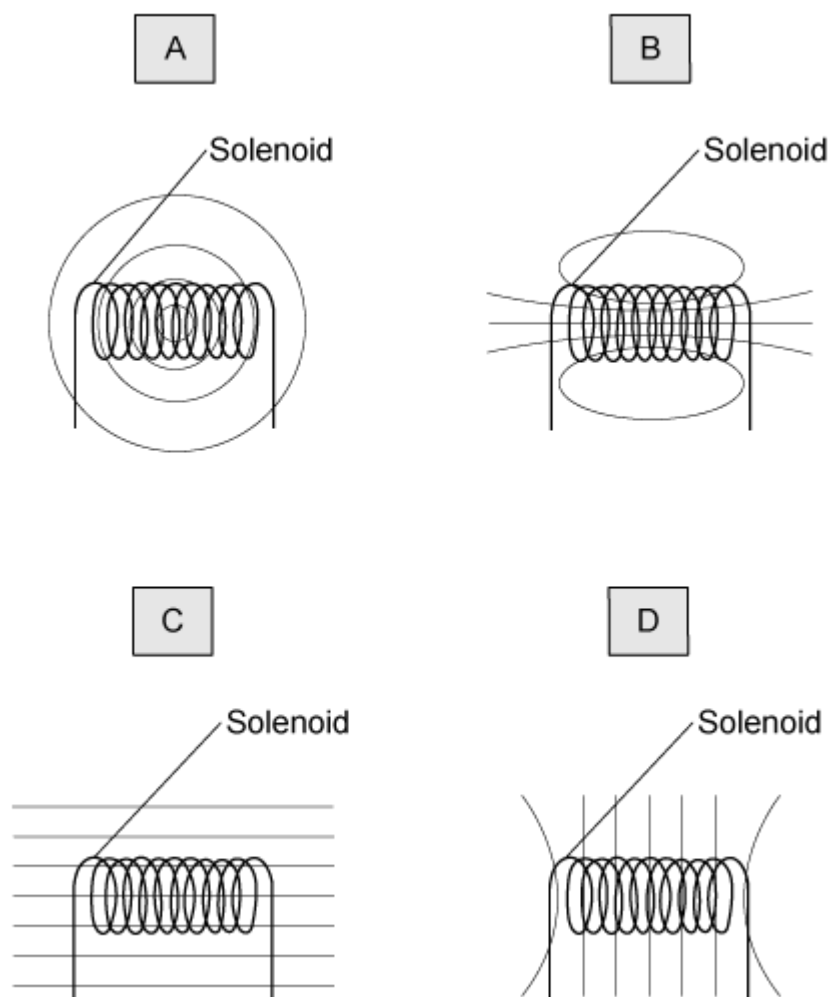
If the input voltage is 48 V, what is the output voltage?

- A.** 2.4 V
- B.** 9.6 V
- C.** 240 V
- D.** 667 V

(1 mark)

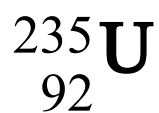
36 Which diagram shows the correct magnetic field pattern around a current-carrying

solenoid?

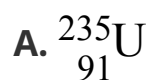


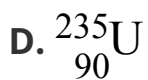
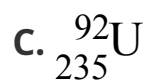
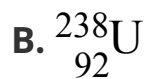
(1 mark)

37 An isotope of uranium is represented by the symbol:



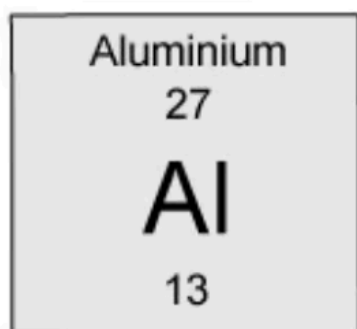
Which of the answers below could be a different isotope of uranium?





(1 mark)

38 How many neutrons are in a nucleus of aluminium? The chemical symbol is given below:



A. 13

B. 27

C. 14

D. 40

(1 mark)

39 A student carried out an experiment to find the half-life of a radioactive substance. Their results are shown in the table below.

Time(seconds)	Count-rate from source(counts per second)
0	300
20	200
40	150
60	100
80	75

What is the half-life of this substance?

- A.** 20 seconds
- B.** 40 seconds
- C.** 60 seconds
- D.** 80 seconds

(1 mark)

40 Which statement about α -particles is correct?

- A.** α -particles consist of two protons and two electrons.
- B.** α -particles are the most penetrating type of radiation.
- C.** α -particles are a type of electromagnetic radiation.
- D.** α -particles are highly ionising.

(1 mark)